

# Modulus, Argument multiplication

## Key Points

The modulus argument form is

$$z = |z|(\cos(\text{Arg}(z)) + i \sin(\text{Arg}(z)))$$

To multiply modulus argument form complex numbers:

- Multiply the moduli,  $|zw| = |z||w|$
- Add the arguments,  $\text{Arg}(zw) = \text{Arg}(z) + \text{Arg}(w)$

To divide use similar rules

Remember to adjust the argument so it is within the range

## Basic Skills

**Q1.** For the following, use the complex numbers,  $z = 9 + 12i$  and  $w = 3 + i$

- Find the modulus, argument form for  $z$  and  $w$ , plotting them on an argand diagram indicating the modulus and argument
- Using the cartesian form multiply the  $z$  and  $w$  to show  $zw = 15 + 45i$
- Express  $zw = 15 + 45i$  in modulus argument, from drawing it on an argand diagram indicating the modulus and argument
- Hence verify the rules  $|zw| = |z||w|$  and  $\text{Arg}(zw) = \text{Arg}(z) + \text{Arg}(w)$

## Competency Skills

**Q2.** Multiply  $z = 2\left(\cos\frac{\pi}{4} + i \sin\frac{\pi}{4}\right)$  and  $w = 5\left(\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right)\right)$

**Q3.** Multiply  $z = \sqrt{3}\left(\cos\frac{\pi}{6} + i \sin\frac{\pi}{6}\right)$  and  $w = 2\sqrt{3}\left(\cos\frac{\pi}{2} + i \sin\frac{\pi}{2}\right)$

**Q4.** By using  $z = x + yi, w = u + vi$  prove the formula  $|zw| = |z||w|$

**Q5. a)** Convert the complex number  $z = \frac{\sqrt{2}}{2}(1 + i)$  into modulus, argument form

**b)** Hence find  $2z\left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right)\right)$

**Q6.** By using  $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$  and  $\text{Arg}\left(\frac{z}{w}\right) = \text{Arg}(z) - \text{Arg}(w)$  calculate the following in modulus, argument form

**a)**  $\frac{2\left(\cos\frac{\pi}{3} + i \sin\frac{\pi}{3}\right)}{5\left(\cos\frac{\pi}{4} + i \sin\frac{\pi}{4}\right)}$

**b)**  $\frac{\sqrt{3}\cos\left(-\frac{2\pi}{3}\right) + \sqrt{3}i \sin\left(-\frac{2\pi}{3}\right)}{\sqrt{12}\cos\frac{2\pi}{3} + \sqrt{12}i \sin\frac{2\pi}{3}}$

**c)**  $\frac{\cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right)}{2i}$

## Advanced Skills

**Q7.** Explain why  $z = |z|(\cos(\text{Arg}(z)) + i \sin(\text{Arg}(z)))$  and  $w = |w|(\cos(\text{Arg}(w)) + i \sin(\text{Arg}(w)))$  have  $zw = |z||w|(\cos(\text{Arg}(z) + \text{Arg}(w)) + i \sin(\text{Arg}(z) + \text{Arg}(w)))$  using an argand diagram

**Q8.** By using  $z = 2\sqrt{3}i - 2$

- Show that  $z = 4\left(\cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3}\right)$
- Find the values for  $z^2, z^3, z^4, z^5$  in modulus argument form
- Hence state the value of  $z^{3n}$  for some constant  $n$
- Hence state a complex solution to the equation  $z^3 = 64$

## Extension

**Q9.** By using  $z = x + yi, w = u + vi$  prove the formula  $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$